

Ali Ghasemzadeh

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🐙 [GitHub](#)

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🌐 [Website](#)

Research Interests

Most of my research focuses on **Computer Vision** and **Medical AI**, where I am particularly interested in developing and understanding learning-based methods for visual and medical data. Alongside this work, I also study theoretical questions in **Estimation Theory**.

Education

Sharif University of Technology (SUT), Tehran, Iran

2022 – Expected Feb 2027

B.Sc. in Computer Engineering | **Minor: Mathematics**

GPA: 18.83/20

National Organization for Development of Exceptional Talents (NODET)

2019 – 2022

High School Diploma in Mathematics and Physics

GPA: 19.5/20

TOEFL iBT (2026 scale): Reading 4.5/6, Listening 5.0/6, Speaking 4.0/6, Writing 4.0/6; Overall 4.5/6.

Research Internship

The University of Hong Kong (HKU), Hong Kong (*In-person*)

Jul 2026 – Sep 2026

Research Intern, Department of Data and Systems Engineering

Supervisor: Prof. Yong-Hong Kuo [🔗](#). Designed an LLM-based pipeline that retrieves and cross-validates industry news, then extracts ontology-constrained knowledge graphs capturing firm tiers and inter-firm relations such as supply and production dependencies.

Publications & Manuscripts

Accepted

PriLoRA: Prior-Conditioned Low-Rank Adapters for Medical Vision Models, Amirhossein Kazerouni* [🔗](#), Sanaz Karimi* [🔗](#), Ali Aghayari* [🔗](#), **Ali Ghasemzadeh***, Reza Azad [🔗](#), Wojciech Samek [🔗](#), Dorit Merhof [🔗](#), Michael Brudno [🔗](#), Babak Taati [🔗](#). **Selected for early acceptance at MICCAI 2026.** Implemented the proposed LoRA variant and baselines across medical image segmentation and classification benchmarks; used empirical results to refine the method and identify suitable priors for the harmonizer. [\[Paper\]](#) [\[Code\]](#)

Toward Accurate RUL and SOH Estimation Using Reinforced Graph-Based PINNs Enhanced with Dynamic Weights, Mohamadreza Akbari Pour, **Ali Ghasemzadeh**, MohamadAli Bijarchi [🔗](#), Mohammad Behshad Shafii [🔗](#). **Accepted for publication in Advanced Engineering Informatics.** Proposed the dynamic-loss component, conducted the literature review, and evaluated the method on NASA C-MAPSS benchmark. [\[arXiv\]](#)

Under Review

A Review of Reward Engineering for Reinforcement Learning Guided LLM Metaheuristic Generation, **Ali Ghasemzadeh***, Armin Khosravi*, Mohamad Mirzadi, Mohamadreza Akbari Pour, Seyedali Mirjalili [🔗](#). **Minor revision at Archives of Computational Methods in Engineering.** Co-led the project; curated and reviewed the literature, developed the taxonomy, and synthesized key challenges and future research directions. [\[SSRN\]](#)

GNBG-C: A Unified and Configurable Benchmark Generator for Constrained Continuous Optimization, **Ali Ghasemzadeh***, Armin Khosravi*, Erfan Babakhani*, Pariya Ghasemzadeh, Yong-Hong Kuo [🔗](#), Seyedali Mirjalili [🔗](#), Amir H. Gandomi [🔗](#). **Submitted to European Journal of Operational Research.** Contributed to the formulation of controllable benchmark parameters; implemented the full generator, authored the methodology and experiments, and visualized the effects of individual control variables. [\[Code\]](#)

Survey on Diffusion Models for Combinatorial Optimization, Armin Khosravi*, **Ali Ghasemzadeh***, Erfan Babakhani*, Amirhossein Kazerouni [🔗](#), Yong-Hong Kuo [🔗](#), Philip Torr [🔗](#), Seyedali Mirjalili [🔗](#). **Submitted to Intelligent Systems with Applications.** Contributed to topic scoping, literature review, taxonomy design, and comparative analysis; authored the taxonomy and comparative-overview sections and revised the manuscript. [\[SSRN\]](#)

In Preparation

Spectral Palette: Learning Diverse Spectral Bases for Implicit Neural Representations, Amirhossein Kazerouni [🔗](#), **Ali Ghasemzadeh**, Armin Khosravi, Donya Jafari, Erfan Babakhani, Ali Diba [🔗](#), Yong-Hong Kuo [🔗](#), Michael Brudno [🔗](#), Babak Taati [🔗](#). **In preparation for submission to CVPR.** Implemented the proposed INR and baselines including SIREN and WIRE; evaluated them on classification, CT reconstruction, image denoising, and super-resolution using benchmarks including Kodak and DIV2K.

Brain Metastasis Segmentation: Loss Function Design, Yousef Sadegheih [🔗](#), Armin Khosravi, **Ali Ghasemzadeh**, Erfan Babakhani, Yong-Hong Kuo [🔗](#), Dorit Merhof [🔗](#). **In preparation for submission to IEEE ISBI.** Implemented baseline losses and a Hungarian-matching loss, conducted comparative experiments, and contributed to manuscript writing for brain metastasis segmentation in MRI.

Multi-Armed Bandit for Adaptive Metaheuristic Selection, Erfan Babakhani*, Armin Khosravi*, **Ali Ghasemzadeh***, Seyedali Mirjalili [🔗](#), Amir H. Gandomi [🔗](#). **Manuscript in preparation.** Contributed to method design and baseline implementation, reviewed related work, and developed comparative visualizations for adaptive metaheuristic selection across heterogeneous tasks.

Learning Under Joint Domain Evolution: Progressive Multi-Source Domain Adaptation , *Ali Sadeghian** , *Ali Ghasemzadeh** , *Seyedreza Tavakoli* , *Amir Naghdi* , *Aria Mokhtari* , *Mahdi Ghiyasi* , *Babak Khalaj*  , *Farzan Farnia*  , *Soheil Kolouri*  , *Ali Diba*  . **In preparation for submission to Transactions on Machine Learning Research (TMLR)**. Co-led the project, formulated the problem, contributed core solution ideas, implemented the proposed method, and coordinated the experiments.

Text-Guided Medical Image Segmentation: A Survey , *Erfan Babakhani** , *Ali Ghasemzadeh** , *Armin Khosravi** , *Pariya Ghasemzadeh* , *Amirhossein Kazerouni*  , *Yong-Hong Kuo*  , *Seyedali Mirjalili*  . **In preparation for submission to Medical Image Analysis**. Contributed to the literature review and paper categorization and co-authored the background and introduction, organizing the methodological landscape of text-guided medical image segmentation. [\[SSRN\]](#)

Learning from Many Minds: A Survey of Multi-Rater Medical Image Segmentation under Annotation Ambiguity , *Erfan Babakhani** , *Armin Khosravi** , *Ali Ghasemzadeh** , *Pariya Ghasemzadeh* , *Sanaz Karimi*  , *Reza Azad*  , *Ulas Bagci*  , *Dorit Merhof*  . **In preparation for submission to Medical Image Analysis**. Contributed to the literature review and paper categorization and co-authored the background and introduction on annotation ambiguity and multi-rater medical image segmentation.

Research Experience

Continual Learning for Out-of-Distribution Data *Jul 2026 – Present*. Advisors: Yousef Sadegheih  , Pratibha Kumari  , Prof. Dorit Merhof  (University of Regensburg) . Trained a 2D nnU-Net for medical image segmentation and contributed to the development and evaluation of the 3D inference pipeline for out-of-distribution data.

Structural Properties of Semidefinite Programs *Nov 2025 – Mar 2026*. Advisor: Prof. Peyman Mohajerin Esfahani  (University of Toronto) . Studied invexity, pseudoconvexity, and hidden convexity in semidefinite programming, focusing on tractable convex reformulations of nonconvex problems.

Honors & Awards

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| • Ranked 1st in Estimation Theory, a graduate-level EE course taken as an undergraduate | 2025 |
| • Ranked 2nd among 60 EE and CE students in Convex Optimization | 2025 |
| • Top 0.03% in the Iranian National University Entrance Exam (Konkour) | 2022 |

Advanced Coursework

AI & ML: Machine Learning, Reinforcement Learning, Generative Models
Math & Optimization: Convex Optimization, Estimation Theory, Stochastic Processes, Mathematical Analysis

Teaching Assistant Experience (Designed theoretical and practical assignments)

Machine Learning Theory: Dr. Najafi (Fall 2025)	Linear Algebra: Dr. Rabiee & Dr. Ramezani (Fall 2024)
Machine Learning: Dr. Motahari (Spring 2025)	Automata Theory: Dr. Dolati (Fall 2024)
Probability & Statistics: Dr. Najafi (Spring & Fall 2024)	Fundamentals of Programming: Dr. Fazli (Fall 2023)

Technical Skills

ML & Scientific Computing: PyTorch, TensorFlow, Hugging Face, PEFT, scikit-learn, NumPy, SciPy, Pandas
Programming & Tools: Python, C/C++, MATLAB, Git, Linux, Docker, $\text{\LaTeX}$
Languages: Persian (Native), English (Professional)